

Practice Test - 08

Lakshya NEET Hindi Test Series (2026)

Physics

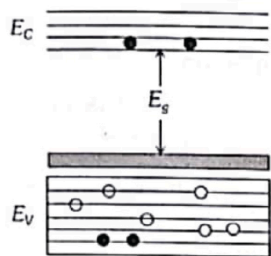
Q1 In a semiconductor, the concentration of electrons is 8×10^{14} per cm^3 and the concentration of holes is 5×10^{12} per cm^3 . Then the semiconductor is:

- (1) p-type (2) n-type
(3) intrinsic (4) PNP type

Q2 In a semiconductor, if the number of holes and conduction electrons are n_p and n_e respectively, then-

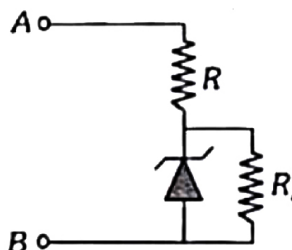
- (1) in a pure semiconductor $n_p > n_e$
(2) in an extrinsic semiconductor $n_p = n_e$
(3) in a pure semiconductor $n_p = n_e$
(4) in a pure semiconductor $n_e > n_p$

Q3 In the energy band diagram of a material shown below, the empty circles and filled circles represent "hole" and electron respectively. Then the material is:



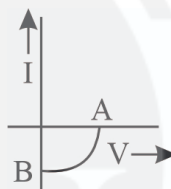
- (1) p-type semiconductor
(2) an insulator
(3) a metal
(4) n-type semiconductor

Q4 If the voltage between terminals A and B is 17 V and the Zener breakdown voltage is 9 V, then the potential difference across R will be:



- (1) 6 V (2) 8 V
(3) 9 V (4) 17 V

Q5 The V-I characteristic curve of a semiconductor device is shown in the figure.



Which of the following statements is **correct** for this

- (1) It is for a 'photodiode' and A and B represent voltage and electric current respectively in open circuit.
(2) It is for an LED and A and B represent voltage in open circuit and electric current in short-circuit respectively.
(3) It is the V-I characteristic curve of a 'solar cell', where A represents the voltage in open circuit and B represents the short-circuit electric current.
(4) It is for a 'solar cell' and A and B represent voltage and electric current respectively in open circuit.

Q6 Match **List-I** with **List-II**. And choose the **correct** option.



	List-I		List-II
(A)	Light Emitting Diode	(I)	Converts light energy into electrical energy.
(B)	Solar cell	(II)	Converts alternating current into direct current.
(C)	Zener diode	(III)	Converts electrical energy into light.
(D)	Rectifier	(IV)	Provides constant voltage.

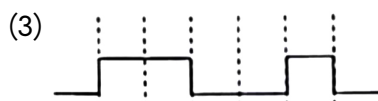
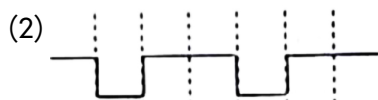
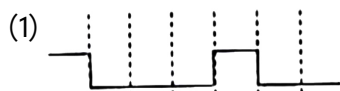
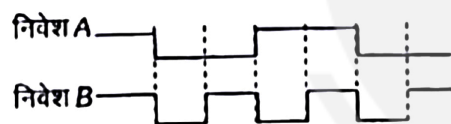
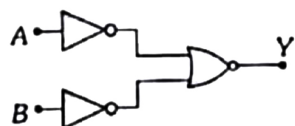
(1) A-III, B-II, C-I, D-IV

(2) A-IV, B-I, C-II, D-III

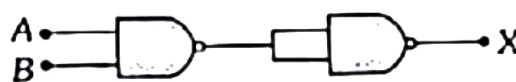
(3) A-III, B-I, C-IV, D-II

(4) A-IV, B-II, C-I, D-III

Q7 The input waveforms 'A' and 'B' for the logic gate shown below are as follows. Select the **correct** output:



Q8 The output (X) of the logic gate (dwar) shown in the diagram will be:



(1) $X = \overline{A + B}$

(2) $X = A + B$

(3) $X = \overline{A \cdot B}$

(4) $X = A \cdot B$

Q9 The Boolean equation for the circuit given in the adjoining figure will be.



(1) $Y = A + \overline{B}$

(2) $Y = \overline{A + B}$

(3) $Y = \overline{A} + B$

(4) $Y = \overline{A} \cdot B$

Q10 Match the symbols shown in **List-I** with their names given in **List-II** and select the **correct** option.

List-I		List-II	
A.		I.	Simple junction diode
B.		II.	Zener diode
C.		III.	Photodiode
D.		IV.	Light emitting diode

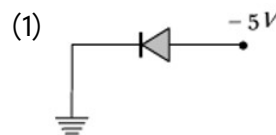
(1) A-I, B-II, C-III, D-IV

(2) A-IV, B-III, C-I, D-II

(3) A-IV, B-III, C-II, D-I

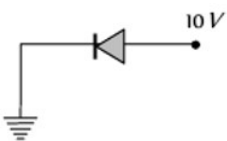
(4) A-II, B-IV, C-III, D-I

Q11 Which of the following semiconductor diodes is in reverse bias



(2)

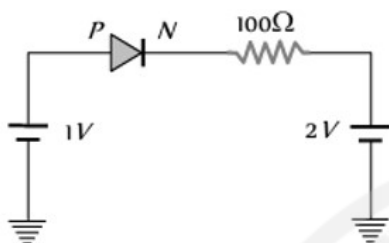




(3) $10V$ $\rightarrow$ $5V$

(4) $-5V$ $\rightarrow$ $-15V$

Q12 The current flowing through an ideal PN-junction in the circuit shown in the figure is



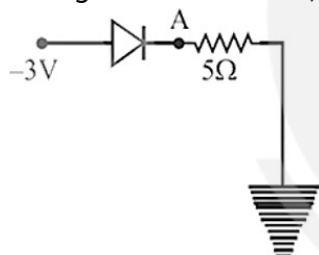
(1) Zero

(2) 1 mA

(3) 10 mA

(4) 30 mA

Q13 In the given figure, the voltage at point A (assuming the diode is ideal) is:



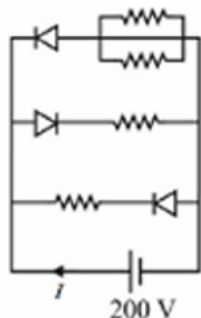
(1) 0 V

(2) -3 V

(3) -2.3 V

(4) -2.7 V

Q14 Calculate the current I in the following circuit, if all diodes are ideal. All resistances are of 200Ω .



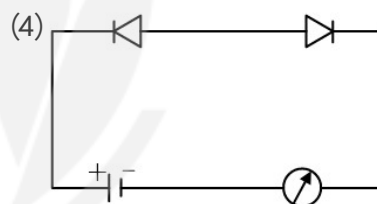
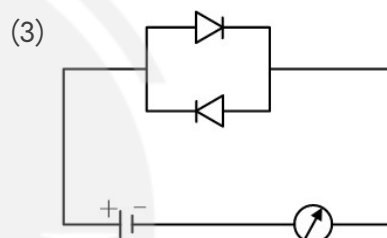
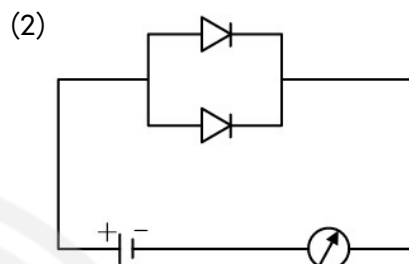
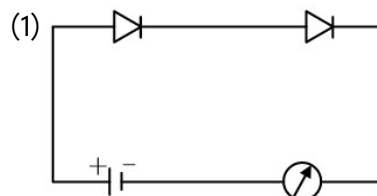
(1) शून्य

(2) 1 A

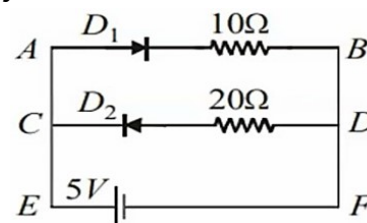
(3) 2 A

(4) 4 A

Q15 Which circuit will not show current in the ammeter?



Q16 Two ideal diodes are connected to a battery as shown in the circuit. The current supplied by the battery is



(1) 0.75 A

(2) शून्य

(3) 0.25 A

(4) 0.5 A

Q17



In a P-type semiconductor, the majority and minority charge carriers are respectively

- (1) protons and electrons
- (2) electrons and protons
- (3) electrons and holes
- (4) holes and electrons

Q18 To make a P-type semiconductor, germanium is doped with

- (1) with boron
- (2) with gallium
- (3) with aluminium
- (4) with all of the above

Q19 At room temperature, a P-type semiconductor has

- (1) a large number of holes (Holes) and a few electrons
- (2) a large number of free electrons and a few holes
- (3) equal number of free electrons and holes
- (4) neither holes nor electrons

Q20 In a sample of (Ge) germanium crystal, Al is added as an impurity. The density of acceptor (Acceptor) atoms is $\sim 10^{21}$ atoms/m³. If in the pure state the density of electron-hole pairs is $\sim 10^{19}$ / m³, then the density of electrons in the sample will be

- (1) $10^{17}/\text{m}^3$
- (2) $10^{15}/\text{m}^3$
- (3) $10^4/\text{m}^3$
- (4) $10^2/\text{m}^3$

Q21 The de Broglie wavelength of a proton accelerated through a potential difference V is. Then, the de Broglie wavelength of an α -particle accelerated through the same potential difference will be

- (1) $\frac{\lambda}{2}$
- (2) $\frac{\lambda}{\sqrt{2}}$

(3) $\frac{\lambda}{2\sqrt{2}}$

(4) $\frac{\lambda}{8}$

Q22 The frequency of the incident photon is ν and the work function is Φ_0 . If the threshold frequency is ν_0 , then the necessary condition for the emission of photoelectrons is:

- (1) $\nu < \nu_0$
- (2) $\nu = \frac{\nu_0}{2}$
- (3) $\nu \geq \nu_0$
- (4) $\nu \leq \phi_0$

Q23 When radiation of wavelength λ is incident on a metallic surface, the stopping potential is 4.8 volts. If the same surface is illuminated with radiation of double the wavelength, the stopping potential becomes 1.6 volts. The threshold wavelength of the surface will be

- (1) 2λ
- (2) 4λ
- (3) 6λ
- (4) 8λ

Q24 The work function of a substance is 4.0 eV. The maximum wavelength of light that can cause emission of photoelectrons from this substance will be:-

- (1) 540 nm
- (2) 400 nm
- (3) 310 nm
- (4) 220 nm

Q25 A 200W sodium bulb emits yellow light of wavelength $0.6 \mu\text{m}$. Assuming that the efficiency of the bulb in converting electrical energy to light is 25%, the number of photons of yellow light emitted per second will be:-

- (1) 1.5×10^{20}
- (2) 6×10^{18}
- (3) 62×10^{20}
- (4) 3×10^{19}

Q26 Regarding photoelectric effect, match **List-I** with **List-II**. And choose the correct option.

	List-I		List-II
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(A)	On increasing intensity of light	(I)	Saturation current decreases
(B)	On increasing frequency of light	(II)	Kinetic energy of emitted photoelectrons increases
(C)	On decreasing intensity of light	(III)	Kinetic energy of emitted photoelectrons decreases
(D)	On decreasing frequency of light	(IV)	Saturation current increases

- (1) (A) - (III), (B) - (II), (C) - (I), (D) - (IV)
 (2) (A) - (IV), (B) - (I), (C) - (II), (D) - (III)
 (3) (A) - (IV), (B) - (III), (C) - (I), (D) - (II)
 (4) (A) - (IV), (B) - (II), (C) - (I), (D) - (III)

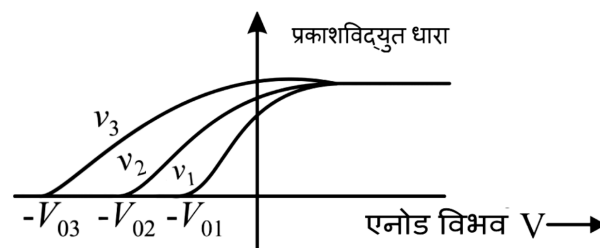
Q27 For the same energy, find the ratio of the wavelengths of a photon and an electron (here m is the mass of the electron)

- (1) $c\sqrt{\frac{2m}{E}}$ (2) $\frac{1}{c}\sqrt{\frac{2m}{E}}$
 (3) $\frac{1}{c^2}\sqrt{\frac{2m}{E}}$ (4) $\frac{1}{c}\sqrt{\frac{2m}{E^2}}$

Q28 The de Broglie wavelength associated with a proton λ , increases by 25%, if its momentum decreases by p_0 then the initial momentum of the proton was:

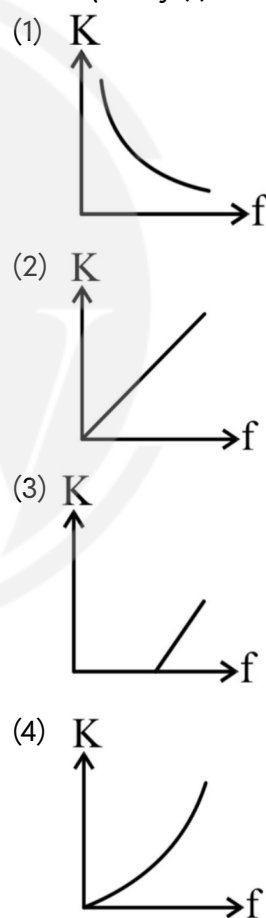
- (1) $4p_0$ (2) $\frac{p_0}{4}$
 (3) $5p_0$ (4) $\frac{p_0}{5}$

Q29 In photoelectric effect, the curve between photoelectric current and anode potential V (for different frequencies) is shown in the figure, then:



- (1) $\nu_1 > \nu_2 > \nu_3$
 (2) $\nu_1 < \nu_2 < \nu_3$
 (3) $\nu_1 = \nu_2 = \nu_3$
 (4) $\nu_1 > \nu_2 < \nu_3$

Q30 Which graph best represents the variation in kinetic energy (K) of photoelectrons as a function of frequency (f) of incident radiation?



Q31 The threshold wavelength of a metal surface is 400 nm. If the maximum kinetic energy of the



emitted photoelectrons is 1.5 eV. Then find the wavelength of the incident photon.

- (1) 2696 Å (2) 2500 Å
(3) 2400 Å (4) 2796 Å

Q32 When a photon of 3.8 eV energy falls on a metallic surface having a work function of 2.8 eV, the maximum kinetic energy of the emitted electrons is:

- (1) 1 eV (2) 6.6 eV
(3) 0 to 1 eV (4) 2.8 eV

Q33 The surface of sodium is illuminated by light of wavelength 6000 Å. The work function of sodium is 1.6 eV. Then the minimum kinetic energy of emitted electrons is:

- (1) 0 eV (2) 1.50 eV
(3) 2.46 eV (4) 4.14 eV

Q34 An α -particle and a carbon-12 atom have the same kinetic energy K . The ratio of their de-Broglie wavelengths ($\lambda_\alpha : \lambda_{C_{12}}$) is

- (1) 1 : $\sqrt{3}$ (2) $\sqrt{3} : 1$
(3) 3 : 1 (4) $\sqrt{2} : 3$

Q35 Two statements are given below:

Statement - I : Two photons having equal linear momenta have equal wavelengths.

Statement - II : If the wavelength of a photon decreases, then the momentum and energy of that photon will also decrease.

In the light of the above statements, choose the **appropriate answer** from the options given below

- (1) Both Statement I and Statement II are correct
(2) Statement I is incorrect but Statement II is correct
(3) Both Statement I and Statement II are incorrect
(4) Statement I is correct but Statement II is incorrect

Q36 In the hydrogen atom, the energy in the n th orbit is $E_n = -\frac{13.6}{n^2} \text{ eV}$ then the energy required to send the electron from the first orbit to the second orbit will be

- (1) 10.2 eV (2) 12.1 eV
(3) 13.6 eV (4) 3.4 eV

Q37 The ionization potential of a hydrogen atom is 13.6 eV. Hydrogen atoms are excited from the ground state by monochromatic radiation of 12.1 eV photon energy. According to Bohr's theory, the spectral lines emitted by the hydrogen atoms will be

- (1) one (2) two
(3) three (4) four

Q38 In a hydrogen atom, the radius of the second orbit of the electron is R . The radius of the third orbit will be

- (1) 3 R (2) 2.25 R
(3) 9 R (4) $R/3$

Q39 In a hydrogen atom, an electron makes a transition from the $n=4$ orbit to the $n=2$ orbit. The wave number of the emitted radiation (where R -Rydberg constant) will be

- (1) $\frac{16}{3R}$ (2) $\frac{2R}{16}$
(3) $\frac{3R}{16}$ (4) $\frac{4R}{16}$

Q40 For the hydrogen atom, match the radiations given in **Column-I** with the series given in **Column-II**.

	Column-I		Column-II
A.	Ultraviolet light	I.	Paschen series
B.	Visible light	II.	None of these
C.	Infrared radiation	III.	Lyman series
D.	Microwave	IV.	Balmer series

- (1) A-IV, B-III, C - II, D - I
(2) A-III, B-IV, C - I, D - II
(3) A-I, B-III, C - IV, D - II



(4) A-III, B-II, C - I, D - IV

- Q41** Fusion reaction takes place at high temperature because:
- (1) nuclei break up at high temperature.
 - (2) atoms are ionised at high temperature.
 - (3) thermal kinetic energy is sufficient to overcome the Coulomb repulsion between the nuclei.
 - (4) molecules break up at high temperature.
- Q42** The radius of ${}_{29}\text{Cu}^{64}$ nucleus will be:
- (1) $1.2 \times 10^{-15} \text{ m}$
 - (2) $2.4 \times 10^{-15} \text{ m}$
 - (3) $3.6 \times 10^{-15} \text{ m}$
 - (4) $4.8 \times 10^{-15} \text{ m}$
- Q43** The binding energy per nucleon of ${}_{3}\text{Li}^7$ and ${}_{2}\text{He}^4$ nuclei are 5.60 MeV and 7.06 MeV respectively. In the nuclear reaction ${}_{3}\text{Li}^7 + {}_{1}\text{H}^1 \rightarrow {}_{2}\text{He}^4 + {}_{2}\text{He}^4$ the value of energy Q released is:
- (1) 19.6 MeV
 - (2) -2.4 MeV
 - (3) 8.4 MeV
 - (4) 17.3 MeV

- Q44** The radius of a Germanium (Ge) nucleus is measured to be twice the radius of ${}_{4}\text{Be}^9$. The number of nucleons in Ge is:
- (1) 72
 - (2) 73
 - (3) 74
 - (4) 75

Q45 Match List-I with List-II:

	List-I		List-II
A.	Nuclear fission	I.	Change in energy it shows that
B.	Nuclear fusion	II.	Nuclear forces are short range.
C.	Mass defect	III.	Atom bomb
D.	Saturation of nuclear force	IV.	Source of sun's energy

Select the **correct** answer from the following options:

- (1) A-(III), B-(II), C-(I), D-(IV)
- (2) A-(IV), B-(I), C-(II), D-(III)
- (3) A-(III), B-(IV), C-(I), D-(II)
- (4) A-(IV), B-(II), C-(I), D-(III)



Answer Key

Q1 (2)
Q2 (3)
Q3 (1)
Q4 (2)
Q5 (3)
Q6 (3)
Q7 (1)
Q8 (4)
Q9 (3)
Q10 (4)
Q11 (1)
Q12 (1)
Q13 (1)
Q14 (2)
Q15 (4)
Q16 (4)
Q17 (4)
Q18 (4)
Q19 (1)
Q20 (1)
Q21 (3)
Q22 (3)
Q23 (2)

Q24 (3)
Q25 (1)
Q26 (4)
Q27 (1)
Q28 (3)
Q29 (2)
Q30 (3)
Q31 (1)
Q32 (1)
Q33 (1)
Q34 (2)
Q35 (4)
Q36 (1)
Q37 (3)
Q38 (2)
Q39 (3)
Q40 (2)
Q41 (3)
Q42 (4)
Q43 (4)
Q44 (1)
Q45 (3)

